STATOTROL* II DRIVES 6VRC2100 AND 6VRC2101 BIDIRECTIONAL REGENERATIVE BRAKING OPTION

DESCRIPTION

The Regenerative Braking Option consists of voltage sensing and switching circuitry and terminals for connecting a remote control station to the Statotrol II controller. The Regenerative Braking Option is not complete in itself, and must be used in conjunction with one of the available reversing options which include the 6VRK1000, 6VRK1001 or 6VRK1002.

The 6VRC2100 is for use on open, chassis mount controllers, and the 6VRC2101 is for use on enclosed controllers.

The Regenerative Braking Option functions by clamping the controller reference signal input to zero, sensing the direction of motor rotation, switching the reversing option to call for rotation in the opposite direction, and then turning on the controller to drive the motor toward stop. When the motor stops, the reference signal (which is clamped to zero) is satisfied, so the controller turns off the power to the armature, and the regenerative braking option then releases.

INSTALLATION

For drives rated 2 HP or less, the reversing option should be installed before the regenerative braking option is installed. The Regenerative Braking Option plugs into the controller in the location normally used for the remote control station adapter. Refer to Figure 2 in the controller instruction book for proper mounting location and orientation. The retained screws at the four corners of the bracket secure the option.

CAUTION

IF A TIMED ACCELERATION OPTION IS TO BE USED, IT MUST HAVE A MAXIMUM ADJUSTABLE TIME OF 8 SECONDS OR LESS. THE 6VTA-1002 TIMED ACCELERATION OPTION IS RECOMMENDED FOR USE WITH THE REGENERATIVE BRAKING OPTION WHEN TIMED ACCELERATION IS REQUIRED.

ELECTRICAL CONNECTIONS

The Regenerative Braking Option and the remote control station must be connected as shown in Figure 1 or Figure 2.

If a motor thermostat is not used, connect the manual override stop switch directly between terminals 54 and 172.

The Forward, Reverse, Brake switch may be a 4 pole, 3 position switch as shown in Figure 1, where F closes only for forward operation, R closes only for reverse operation, one Brake contact opens only for braking, and the other Brake contact closes only for braking. The forward, reverse, brake switching may also be done by a pair of 3 pole relays connected as shown in Figure 2.

A tachometer or any other suitable signal source may be used in place of the motor armature shown in Figures 1 and 2. If motor armature voltage is to be used as the signal input to the regenerative brake, make the electrical connections for the signal at controller terminals A1 and A2.

INITIAL OPERATION AND ADJUSTMENTS

CAUTION

IF THE MINIMUM SPEED ADJUSTMENT ON THE CONTROLLER MAIN COMPONENT BOARD IS MISADJUSTED, THE MOTOR MAY OSCILLATE FROM FORWARD TO REVERSE WHEN THE OPERATOR'S SWITCH IS SET TO "BRAKE".

CAUTION

SWITCHING THE OPERATOR'S CONTROL TO "BRAKE" WHILE THE MOTOR IS IN AN OVERSPEED CONDITION MAY RESULT IN FUSE BLOWING OR SCR FAILURE. SET THE MAXIMUM SPEED ADJUSTMENT ON THE CONTROLLER MAIN COMPONENT BOARD TO PREVENT OVERSPEED.

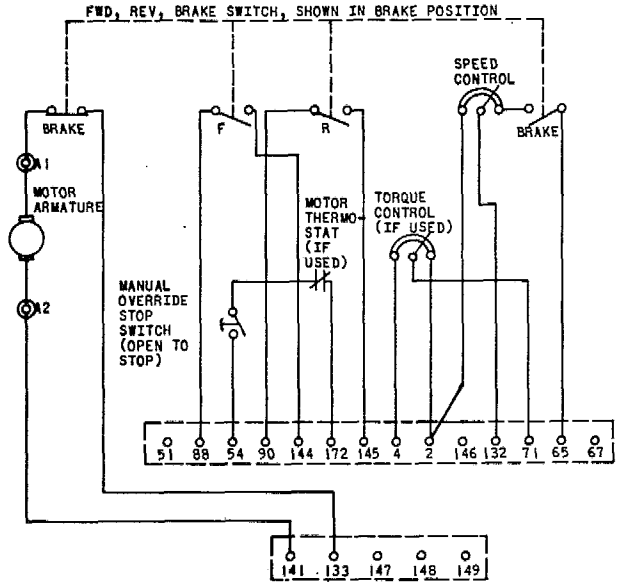
After completing the prepower checks and adjustments given in the controller instruction book, apply power to the controller, turn the speed control knob to zero, and press the forward or reverse button. If the motor turns, rotate the minimum speed adjustment (on the controller main component board) CCW until the motor stops. Then slowly turn the speed control knob until the motor starts

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company

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to run. Check the direction of motor rotation. (If the motor turns the wrong way, disconnect power from the controller and reverse the motor armature leads.) Next turn the speed control knob until the motor is running at about 1/4 to 1/2 of rated speed, and move the operator's switch to "brake". If the motor oscillates back and forth, the minimum speed adjustment is turned too far clockwise. If the motor coasts to a stop, turn the minimum speed adjustment a few degrees clockwise, and again start the motor and then move the operators switch to "brake". Repeat this process, slowly advancing the minimum speed adjustment, until the motor brakes to a smooth complete stop.

The Statotrol II controller instruction book states that the minimum speed adjustment is normally used to set a minimum speed at which the motor will run with the speed control knob turned to zero. However, when regenerative braking is used, the minimum speed adjustment cannot be used for this purpose. When the application requires that the motor run at some minimum RPM when the speed control knob is set to zero, connect a speed trim potentiometer in series with the speed control potentiometer. Connect the wiper of the speed trim potentiometer to circuit 2 and connect one end of the speed trim potentiometer to the speed control potentiometer. Increasing the resistance in series with the speed control potentiometer will increase the speed at which the motor runs when the speed control potentiometer is set to zero.



NOTE 1: TERMINAL A1 IS POSITIVE WITH RESPECT TO A2 WHEN MOTOR IS RUNNING WITH CONTACT F CLOSED.

2: A MANUAL OVERRIDE STOP SWITCH MUST BE CONNECTED AS SHOWN, SINCE COMPONENT FAILURE CAN PREVENT THE NORMAL SWITCHING ARRANGEMENT FROM STOPPING THE MOTOR.

Figure 1. Remote Control Station Connection Diagram for Control by Manual Switching

CAUTION

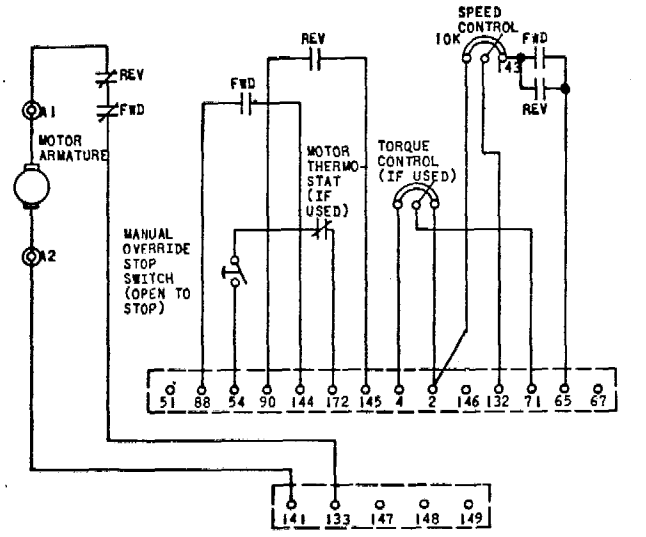
THE MAXIMUM SPEED ADJUSTMENT MUST BE RESET AFTER THE SPEED TRIM POTENTIOMETER IS ADJUSTED.

Next turn the speed control knob to full speed and turn the maximum speed adjustment potentiometer on the controller main component board until the motor runs at the highest speed required for the application. Do not exceed rated speed.

CAUTION

DO NOT MOVE THE OPERATOR'S SWITCH TO THE BRAKE POSITION WHILE THE MOTOR IS RUNNING FASTER THAN RATED SPEED. IF A TACHOMETER IS NOT READILY AVAILABLE FOR CHECKING MOTOR SPEED, TURN THE SPEED CONTROL KNOB TO FULL SPEED, MEASURE THE DC AVERAGE ARMATURE VOLTAGE WITH A MULTIMETER, AND ADJUST THE MAXIMUM SPEED ADJUSTMENT UNTIL THE DC ARMATURE VOLTAGE IS EQUAL TO OR LESS THAN THE MOTOR NAMEPLATE RATED VALUE.

Refer to the factory for special instructions when the Regenerative Braking Option is to be used on a control which uses a remote accessory adapter option and is connected as an instrument follower or tachometer follower.



NOTE 1: FWD AND REV ARE CUSTOMER'S RELAYS. PICK UP FWD TO RUN FORWARD, PICK UP REV TO RUN REVERSE, DROP OUT BOTH TO BRAKE.

2: TERMINAL A1 IS POSITIVE WHEN RESPECT TO A2 WHEN MOTOR IS RUNNING WITH RELAY FWD PICKED UP.

3: A MANUAL OVERRIDE STOP SWITCH MUST BE CONNECTED AS SHOWN, SINCE COMPONENT FAILURE CAN PREVENT THE NORMAL SWITCHING ARRANGEMENT FROM STOPPING THE MOTOR.

Figure 2. Remote Control Station Connection Diagram for Control by Relay Switching

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PRINTED IN U.S.A.

GEK-47610 (7/75) 1M